Warren truss

Plane truss — analysis and CIRSOC 301 rod checks

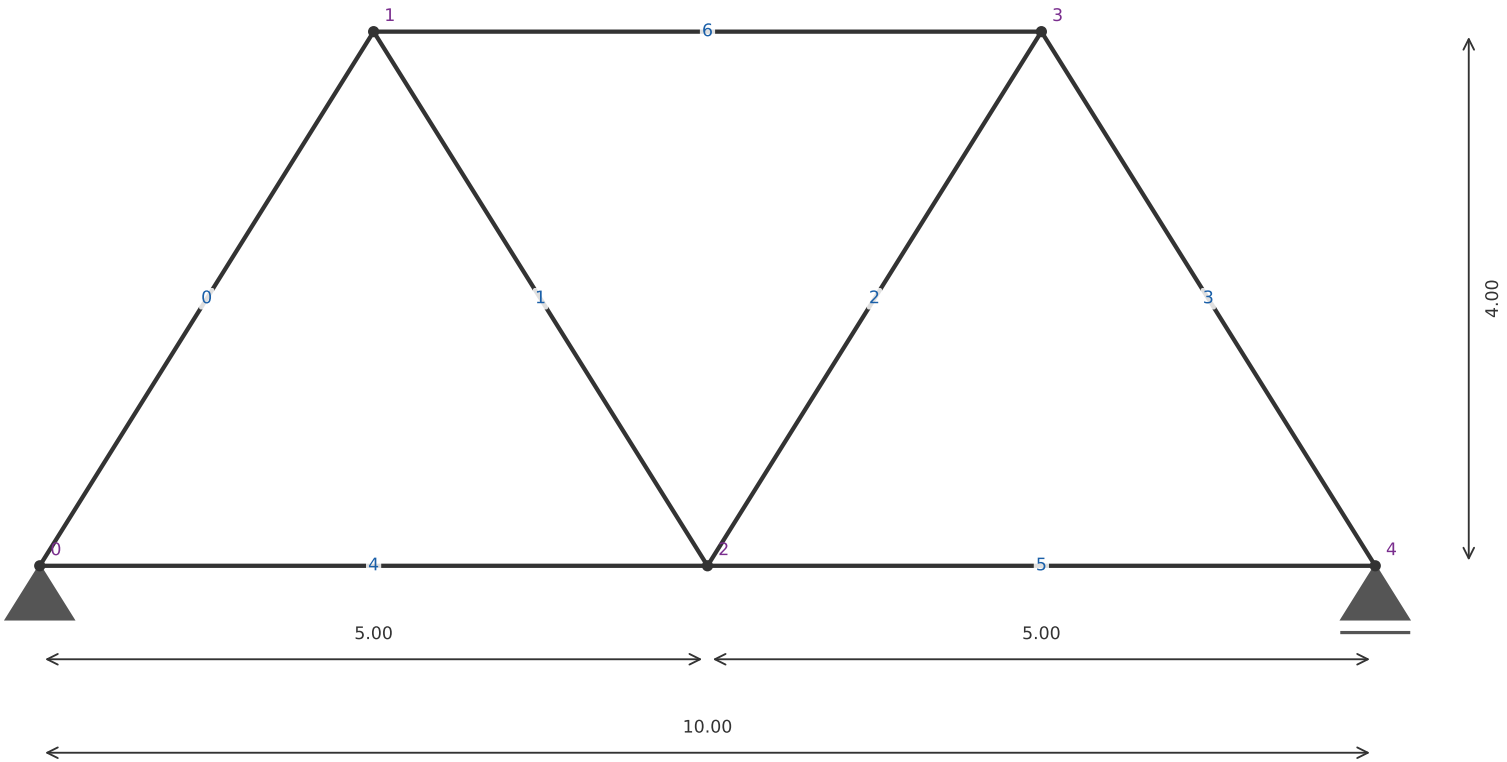
Span	10.00 m
Height	4.00 m
Nodes / rods	5 / 7
Supports	pin at 0, rollerX at 4
Loads	3 nodal, total 120.0 kN
Family "Chord"	CHS 88.9x4.0 (F24 (CIRSOC) / A36)
Family "Web"	CHS 76.1x3.6 (F24 (CIRSOC) / A36)
Weight	Weighs 252 kg, carries 120.0 kN (49 × its own weight), FAILS: 2 rod(s) over capacity.
Deflection	4.97 mm = L/2011, limit L/300: OK
Stability	Statically determinate (m + r − 2j = 7 + 3 − 2×5 = 0): statics alone gives every force.
Most used rod	rod 0 at 1.87 (elastic buckling (E3-3)) — OVER CAPACITY
Rods over capacity	2 of 7 checked
Buckling	λ_1 = 0.68 — BELOW 1, it buckles before the full load is on
Shear and moment	none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

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3. Axial forces
4. Deflection and stress
5. Reactions
6. Utilisation
7. Buckling
8. Rods
9. Explain rod 0 and 3

Node numbers in purple, rod numbers in blue. Dimensions in m.



2 m · true to scale

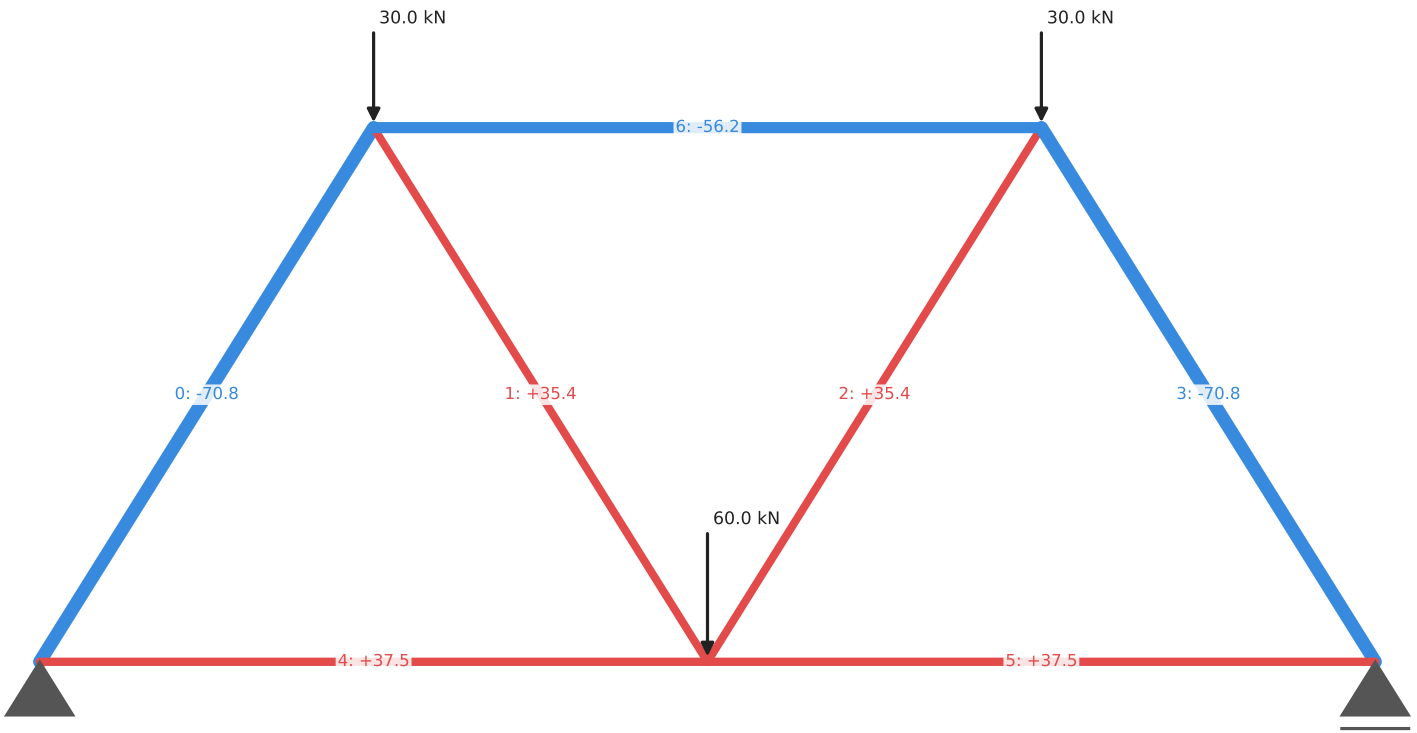
0 2

Truss — Geometry

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MODEL	SIZE	UNITS	CODE	DATE
Warren truss	5 nodes / 7 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 70.8).



2 m · true to scale

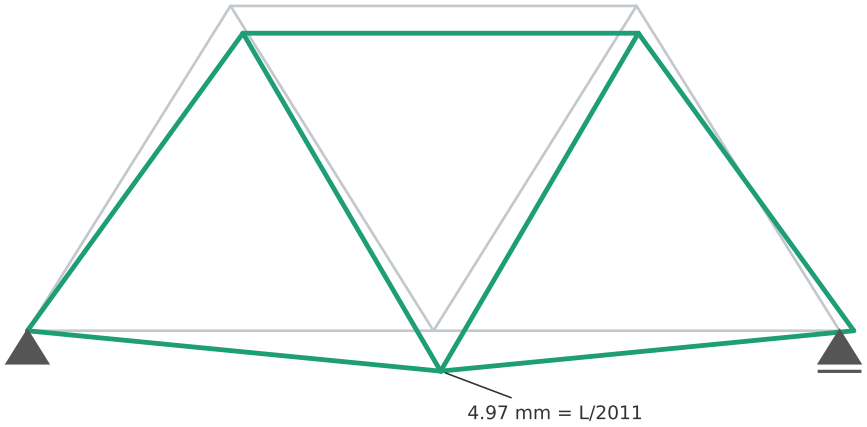
0 2

Truss — Axial forces

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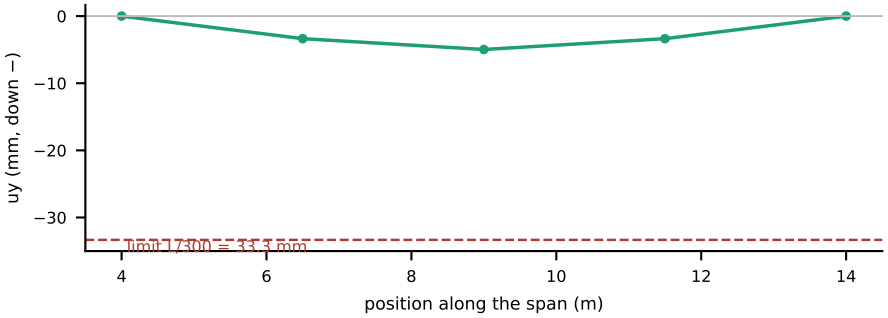
MODEL	SIZE	UNITS	CODE	DATE
Warren truss	5 nodes / 7 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Deflected shape (green) over the truss (grey). Geometry to scale; displacements magnified × 100. Largest vertical deflection 4.97 mm against the limit L/300 = 33.3 mm: OK.



Elastic deformation

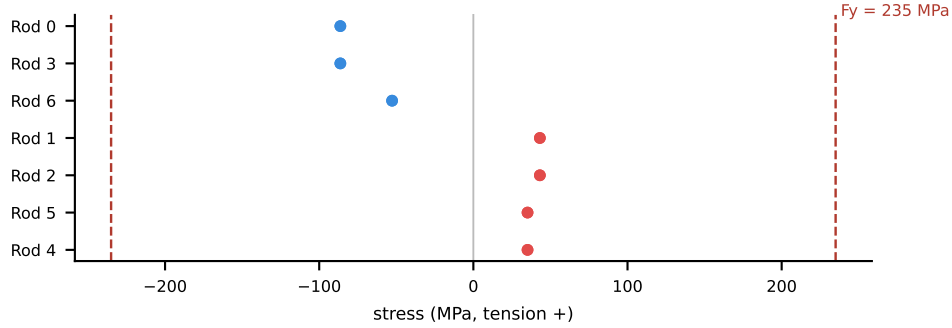
vertical movement of every joint



Largest 4.97 mm at joint 2 = L/2011. Within the limit.

Fibre stress

N/A ± M·c/I at each rod's extreme fibres



Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

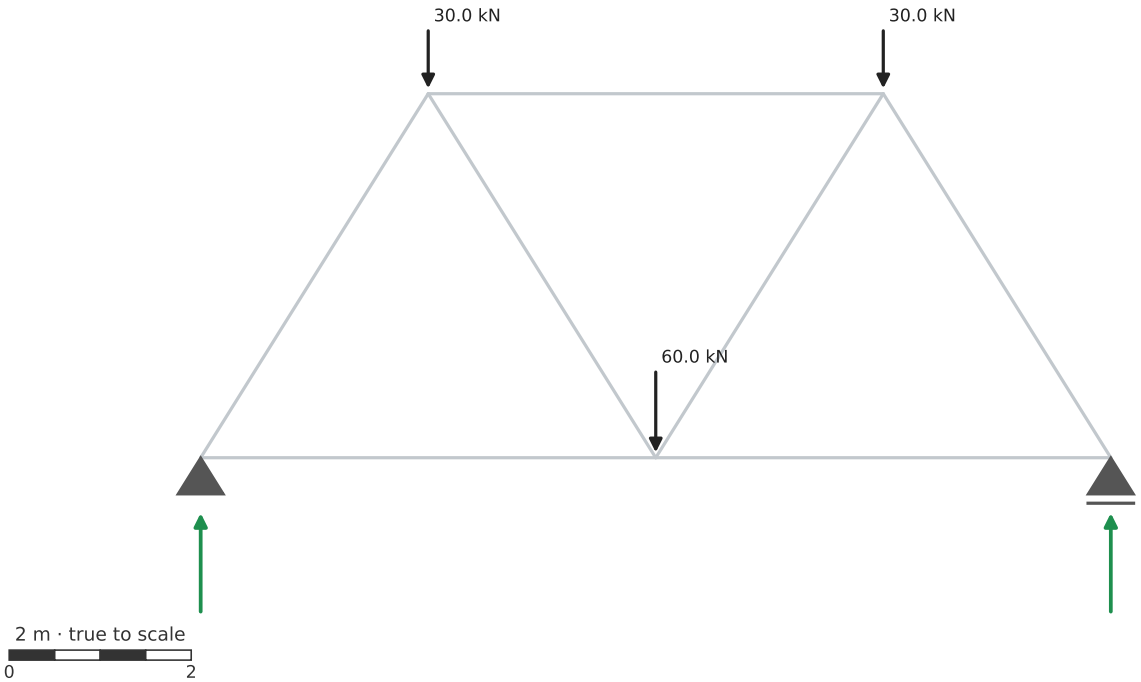
Truss — Deflection and stress

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MODEL	SIZE	UNITS	CODE	DATE
Warren truss	5 nodes / 7 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

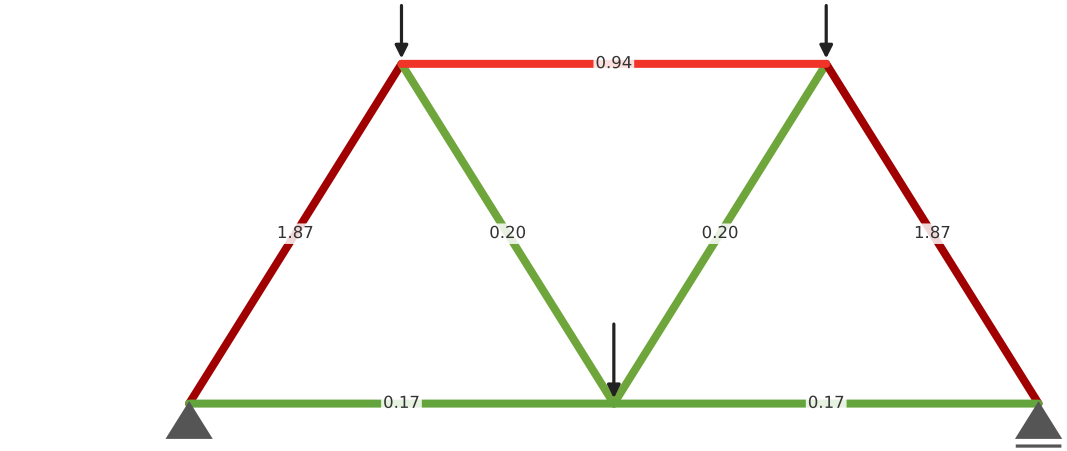
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	60.00
4	rollerX	0.00	60.00
Σ		0.00	120.00



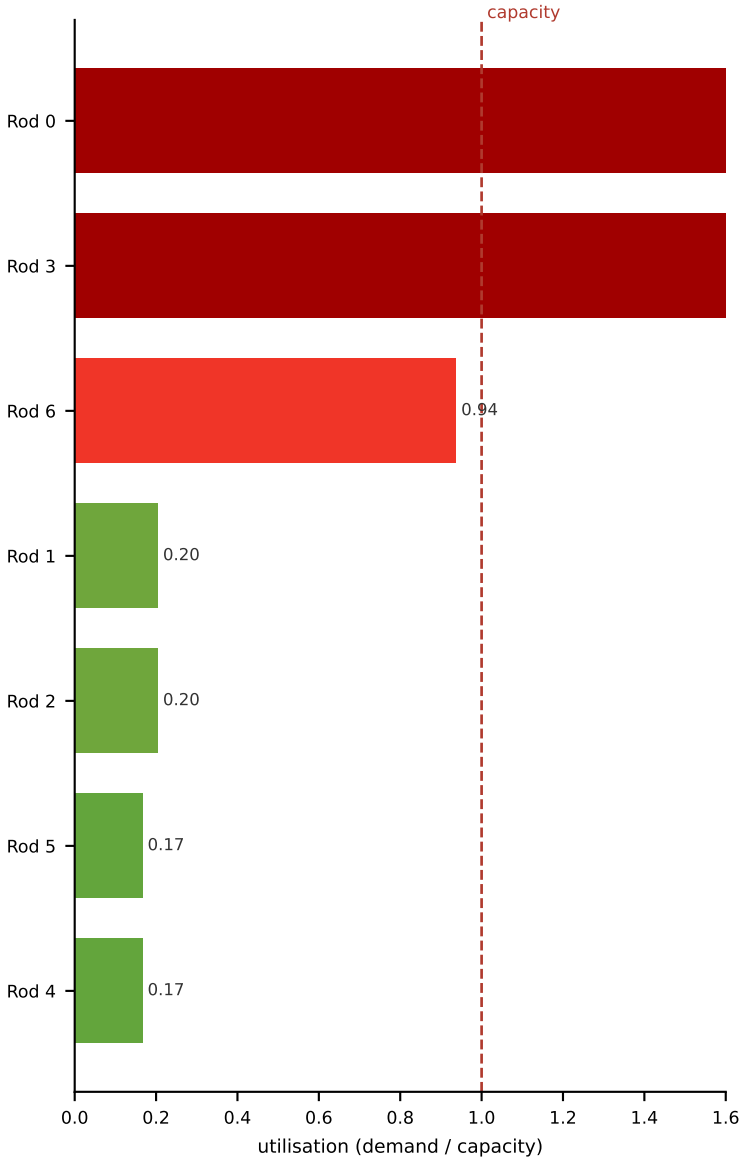
Truss — Reactions

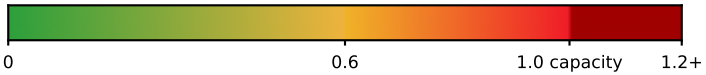
MODEL	SIZE	UNITS	CODE	DATE
Warren truss	5 nodes / 7 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Utilisation: demand over CIRSOC 301 capacity. Most used rod 0 at 1.87 (elastic buckling (E3-3)). 2 rod(s) OVER capacity: 0, 3.



2 m · true to scale
0 2





grey = no steel section, not checked

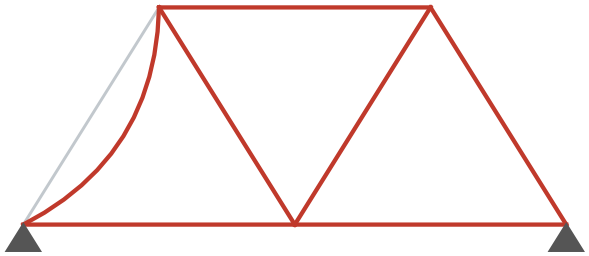
Truss — Utilisation

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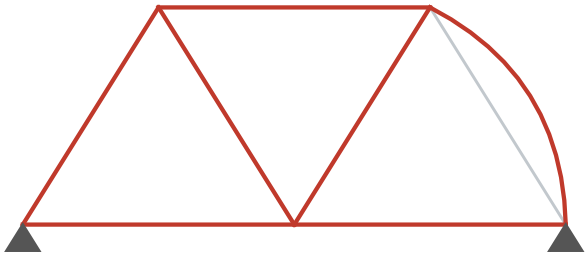
MODEL	SIZE	UNITS	CODE	DATE
Warren truss	5 nodes / 7 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

λ is the factor on the load as it stands. $\lambda_1 = 0.68$ is BELOW 1: it buckles before the full load is on. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

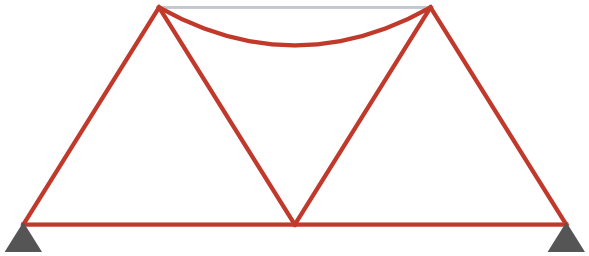
Mode 1: λ = 0.68
rod 0 bows

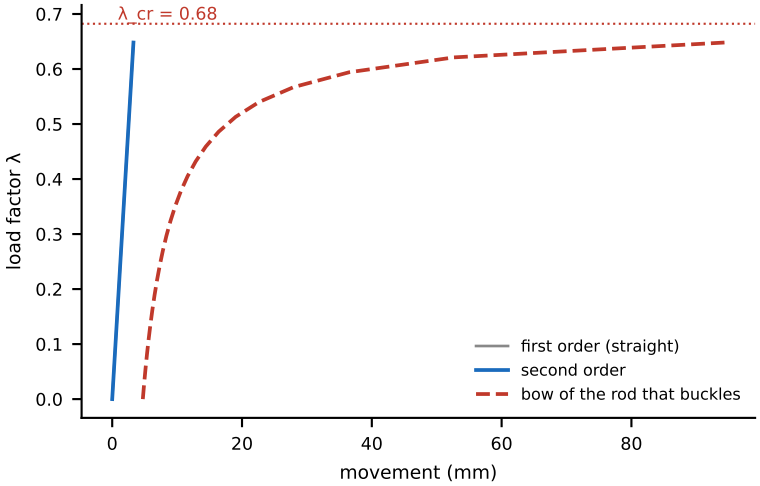


Mode 2: λ = 0.68
rod 3 bows



Mode 3: λ = 1.36
rod 6 bows





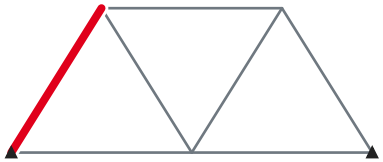
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 4.72 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Truss — Buckling

MODEL	SIZE	UNITS	CODE	DATE
Warren truss	5 nodes / 7 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use	
0	0–1	Web	CHS 76.1x3.6	4.72	-70.75	elastic buckling (E3-3)	1.87	OVER
1	1–2	Web	CHS 76.1x3.6	4.72	+35.38	normal stress (H.3.4)	0.20	OK
2	2–3	Web	CHS 76.1x3.6	4.72	+35.38	normal stress (H.3.4)	0.20	OK
3	3–4	Web	CHS 76.1x3.6	4.72	-70.75	elastic buckling (E3-3)	1.87	OVER
4	0–2	Chord	CHS 88.9x4.0	5.00	+37.50	normal stress (H.3.4)	0.17	OK
5	2–4	Chord	CHS 88.9x4.0	5.00	+37.50	normal stress (H.3.4)	0.17	OK
6	1–3	Chord	CHS 88.9x4.0	5.00	-56.25	elastic buckling (E3-3)	0.94	OK



Rod 0

utilisation 1.87 OVER

elastic buckling (E3-3)

```
Rod 0
=====
Steel, CIRSOC 301-2018 -- A = 820 mm², Fy = 235 MPa, L = 4.717 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,717 mm / 25.7 mm
      = 183.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 183.8²
      = 58.43 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 183.8 vs 4.71 √(200,000/235) = 137.4
      = elastic

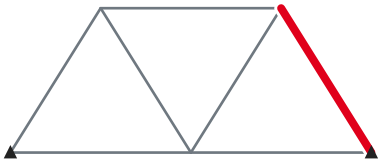
 4. Critical stress  [E3-3]
      Fcr = 0.877 · Fe
      = 0.877 × 58.43
      = 51.25 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 51.25 × 820
      = 37.82 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 70.75 / 37.82
      = 1.871

 7. Governs
      the largest ratio
      = elastic buckling (E3-3)
      = 1.871

utilisation 1.871 -- OVER CAPACITY
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Rod 3

utilisation 1.87 OVER

elastic buckling (E3-3)

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Rod 3
=====
Steel, CIRSOC 301-2018 -- A = 820 mm², Fy = 235 MPa, L = 4.717 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,717 mm / 25.7 mm
      = 183.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 183.8²
      = 58.43 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 183.8 vs 4.71 √(200,000/235) = 137.4
      = elastic

 4. Critical stress  [E3-3]
      Fcr = 0.877 · Fe
      = 0.877 × 58.43
      = 51.25 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 51.25 × 820
      = 37.82 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 70.75 / 37.82
      = 1.871

 7. Governs
      the largest ratio
      = elastic buckling (E3-3)
      = 1.871

utilisation 1.871 -- OVER CAPACITY
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Truss — Explain rod 0 and 3

MODEL	SIZE	UNITS	CODE	DATE
Warren truss	5 nodes / 7 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58